H.W-3 CHE-321 Parthav Patel UIN:655437221

P4-4$_B$ Stoichiometry. The elementary gas reaction

$$2A + B \rightarrow C$$

is carried out isothermally in a PFR with no pressure drop. The feed is equal molar in A and B, and the entering concentration of A is 0.1 mol/dm^3. Set up a stoichiometric table and then determine the following.

1. What is the entering concentration (mol/dm^3) of B?
2. What are the concentrations of A and C (mol/dm^3) at 25% conversion of A?
3. What is the concentration of B (mol/dm^3) at 25% conversion of A? (**Ans:** $C_B = 0.1$ mol/dm^3)
4. What is the concentration of B (mol/dm^3) at 100% conversion of A?
5. If at a particular conversion the rate of formation of C is 2 mol/min/dm^3, what is the rate of formation of A at the same conversion?
6. Write $-r_A$ solely as a function of conversion (i.e., evaluating all symbols) when the reaction is an elementary, irreversible, gas-phase, isothermal reaction with no pressure drop with an equal molar feed and with $C_{A0} = 2.0$ mol/dm^3 at, $k_A = 2$ dm^6/mol•s.
7. What is the rate of reaction at $X = 0.5$?

1.

given $C_{A_0} = 0.1 \frac{mol}{dm^3}$

The feed is equimolar in A and B

Thus, $C_{A_0} = C_{B_0} = 0.1 \frac{mol}{dm^3}$

2. $2A + B \rightarrow C$ $\quad X = 0.25$

$C_A = C_{A_0}(1-X)$

$C_A = 0.1 \frac{mol}{dm^3}(1-0.25)$

$$\boxed{C_A = 0.075 \frac{mol}{dm^3}}$$

$C_C = ?$ at $X = 0.25$

$$\frac{C_C - C_{C0}}{\nu_C} = \frac{C_A - C_{A0}}{\nu_A}$$

$C_{C0} = 0$, using $\nu_C = +1$, $\nu_A = -2$

$$C_C = -\frac{\nu_C}{\nu_A}\left(C_A - C_{A0}\right)$$

$$= -\frac{1}{-2}\left(C_{A0}(1-X) - C_{A0}\right)$$

$$= \frac{1}{2} C_{A0} X$$

$$C_C = \frac{1}{2} \times 0.1 \frac{mol}{dm^3} \times 0.25$$

$$\boxed{C_C = 0.0125 \frac{mol}{dm^3}}$$

3.

$$C_B = C_{B0} - C_{A0} X \left(\frac{\nu_B}{|\nu_A|}\right)$$

$C_{B0} = C_{A0} = 0.1 \frac{mol}{dm^3}$, $\nu_B = -1$, $|\nu_A| = 2$

C_B at $X = 0.25$

$$C_B = 0.1 - 0.1 \times 0.25 \times \left(\frac{1}{2}\right)$$

$$C_B = 0.1 - 0.0125$$

$$\boxed{C_B = 0.0875 \frac{mol}{dm^3}}$$

4.
$$C_B = C_{B_0} - C_{A_0} X \left(\frac{|\nu_B|}{|\nu_A|} \right)$$

C_B, at $X = 1$

$$C_B = 0.1 - 0.1 \times 1 \times \frac{1}{2}$$

$$C_B = 0.1 - 0.05$$

$$\boxed{C_B = 0.05 \ \frac{mol}{dm^3}}$$

5.
$$\frac{r_A}{\nu_A} = \frac{r_C}{\nu_C} \qquad \nu_A = -2$$

$$\nu_C = +1$$

$$r_A = r_C \times \left(\frac{\nu_A}{\nu_C} \right)$$

$$r_A = \frac{2 \ mol}{\frac{min}{dm^3}} \times \left(\frac{-2}{1} \right)$$

$$\boxed{r_A = -4 \ \frac{mol}{\frac{min}{dm^3}}}$$

6.
$$-r_A = k_A C_A^2 C_B \qquad C_A = C_{A_0}(1-X)$$

$$C_B = C_{B_0} - C_{A_0} X \left(\frac{|\nu_B|}{|\nu_A|} \right)$$

$$= C_{A_0} - C_{A_0} x \left(\frac{1}{2}\right) = C_{A_0}\left(1 - \frac{x}{2}\right)$$

$$-r_A = k_A [C_{A_0}(1-x)]^2 \left[C_{A_0}\left(1-\frac{x}{2}\right)\right]$$

$$-r_A = k_A C_{A_0}^3 (1-x)^2 \left(1-\frac{x}{2}\right)$$

$$-r_A = 2 \times (2.0)^3 \times (1-x)^2 \left(1-\frac{x}{2}\right)$$

$$= 2 \times 8 \times (1-x)^2 \left(1-\frac{x}{2}\right)$$

?

?

$$\boxed{r_A = -16 (1-x)^2 \left(1-\frac{x}{2}\right)}$$

? ?

7.

$$-r_A = 16 (1-x)^2 \left(1 - \frac{x}{2}\right)$$

at $x = 0.5$

$$-r_A = 16 (1-0.5)^2 \left(1 - \frac{0.5}{2}\right)$$

$$= 16 \times 0.25 \times 0.75$$

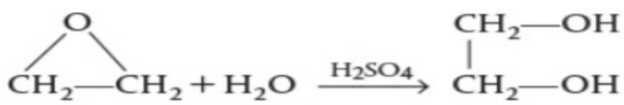
$$= 4 \times 0.75$$

$$-r_A = 3 \frac{mol}{dm^3 . s}$$

$$\boxed{r_A = 3 \frac{mol}{dm^3 . s}}$$

P4-5$_A$ Set up a stoichiometric table for each of the following reactions and express the concentration of each species in the reaction as a function of conversion, evaluating all constants (e.g., =, =). Next, assume the reaction follows an elementary rate law, and write the reaction rate solely as a function of conversion, that is, $-r_A = f(X)$.

1. For the liquid-phase reaction



the entering concentrations of ethylene oxide and water, after mixing the inlet streams, are 16.13 and 55.5 mol/dm^3, respectively. The specific reaction rate is $k = 0.1$ dm^3/mol · s at 300 K with $E = 12500$ cal/mol.

1. After finding $-r_A = f(X)$, calculate the CSTR space-time, τ, for 90% conversion at 300 K and also at 350 K.
2. If the volumetric flow rate is 200 liters per second, what are the corresponding reactor volumes? (**Ans:** At 300 K: $V = 439$ dm^3 and at 350 K: $V = 22$ dm^3)

Given $C_{A_0} = 16.13 \dfrac{mol}{dm3}$

$C_{B_0} = 55.5 \dfrac{mol}{dm3}$

$K_{300} = 0.1 \dfrac{dm^3}{mol \cdot S}$ $\quad E = 12500 \dfrac{cal}{mol}$

$R = 1.987 \dfrac{cal}{mol \cdot K}$

$X = 0.90$

$-r_A = K C_A C_B$ Rate law

$-r_A = K[C_{A_0}(1-X)][C_{B_0} - C_{A_0}X]$

CSTR equation.

$$\tau = \frac{C_{A_0} X}{-r_A}\Big|_X$$

$$\tau = \frac{C_{A_0} X}{K C_{A_0}(1-X)(C_{B_0} - C_{A_0}X)}$$

Cancel C_{A_0}

$$\tau = \frac{X}{K(1-X)(C_{B_0} - C_{A_0}X)}$$

$X = 0.9$
$K = 0.1$

$C_{B_0} - C_{A_0}X = 55.5 - (16.13)(0.9)$
$= 40.98$

$$\tau_{300} = \frac{0.9}{0.1(0.1)(40.98)} = \frac{0.9}{0.4098}$$

$$\boxed{\tau_{300} = 2.20\,s}$$

Q1 Answer.

Rate constant at 350K

$$\ln\left(\frac{k_{350}}{k_{300}}\right) = -\frac{E}{R}\left(\frac{1}{350} - \frac{1}{300}\right)$$

$$\ln\left(\frac{k_{350}}{0.1}\right) = \frac{-12500}{1.987}(-0.000476)$$

$$\ln\left(\frac{k_{350}}{0.1}\right) = 2.996$$

$$k_{350} = 0.1\, e^{2.996}$$

$$k_{350} \simeq 2.0 \frac{dm^3}{mol \cdot s}$$

Space time at 350 K

$$\tau_{350} = \frac{0.9}{(2.0)(0.1)(40.98)}$$

$$\boxed{\tau_{350} = 0.11 s}$$

$\Rightarrow$ Reactor Volume $V = \tau \dot{v}$

Q2 $V_{300} = (2.20)(200) = \boxed{439\, dm^3}$

Answer. $V_{350} = (0.11)(200) = \boxed{22\, dm^3}$

P4-3$_A$ The elementary reversible reaction

$$2A \rightleftarrows B$$

is carried out in a flow reactor where pure A is fed at a concentration of 4.0 mol/dm^3. If the equilibrium conversion is found to be 60%,

1. What is the equilibrium constant, K_C if the reaction is a gas-phase reaction? (**Ans:** Gas: $K_C = 0.328$ dm^3/mol)
2. What is the K_C if the reaction is a liquid-phase reaction? (**Ans:** Liquid: $K_C = 0.469$ dm^3/mol)

3. Write $=r_A$ solely as a function of conversion (i.e., evaluating all symbols) when the reaction is an elementary, reversible, gas-phase, isothermal reaction with no pressure drop with $k_A = 2$ dm^6/mol•s and $K_C = 0.5$ all in proper units.
4. Repeat (c) for a constant-volume batch reactor.

1) $2A \rightleftarrows B$

$C_{A_0} = 4 \frac{mol}{dm3}$

$X_e = 0.6$

$$C_i = \frac{C_{A_0} \frac{F_i}{F_{A_0}}}{1+\varepsilon X}$$

$\varepsilon = y_{A_0} \delta$

$y_{A_0} = 1$

$\varepsilon = -0.5$

$F_A = F_{A_0}(1-X)$

$F_B = F_{A_0} \frac{1}{2} X$

$\delta = \frac{1 \text{ mol } B}{-2 \text{ mol } A} = 0.5$

$$C_A = C_{A_0}\frac{1-X_e}{1+\varepsilon X_e} = \frac{4\ mol}{dm^3}\left(\frac{1-0.6}{1-0.5(0.6)}\right)$$

$$= \frac{1.6}{0.7}\ \frac{mol}{dm^3}$$

$$C_B = C_{A_0}\frac{\frac{1}{2}X_e}{1+\varepsilon X_e} = \frac{0.5(0.6)}{1-0.5(0.6)} = \frac{1.2}{0.7}\frac{mol}{dm^3}$$

$$K_c = \frac{C_B}{C_A^2} = \frac{\frac{1.2}{0.7}\frac{mol}{dm^3}}{\left(\frac{1.6\ mol}{0.7\ dm^3}\right)^2} = \boxed{0.328125\ \frac{dm^3}{mol}}$$

2)

$$C_i = C_{A_0}\frac{F_i}{F_{A_0}}$$

$$C_A = C_{A_0}(1-X_e) = 4\frac{mol}{dm^3} \times (1-0.6)$$

$$= 1.6\ \frac{mol}{dm^3}$$

$$C_B = C_{A_0}\frac{1}{2}X_e = 4\frac{mol}{dm^3} \times \frac{1}{2} \times 0.6$$

$$= 1.2\ \frac{mol}{dm^3}$$

$$K_c = \frac{C_B}{C_A^2} = \frac{1.2\frac{mol}{dm^3}}{\left(1.6\frac{mol}{dm^3}\right)^2} = \frac{1.2}{2.56}\frac{dm^3}{mol}$$

$$K_C \approx 0.46875 \frac{dm^3}{mol}$$

3.

$$-r_A = K_A[A]^2 - k_{-B}[B]$$

$$K_C = 0.5 \frac{dm^3}{mol}$$

$$K_C = \frac{k_A}{k_{-B}} \Rightarrow k_{-B} = \frac{k_A}{K_C} = \frac{2\,dm^6/(mol^2 \cdot s)}{0.5 \frac{dm^3}{mol}}$$

$$= 4 \frac{dm^3}{mol \cdot s}$$

$$[A] = C_{A_0}\left(\frac{1-X}{1+\varepsilon X}\right)$$

$$[B] = C_{A_0}\left(\frac{0.5X}{1+\varepsilon X}\right)$$

$$C_{A_0} = 4 \frac{mol}{dm^3} \qquad \varepsilon = -0.5$$

$$-r_A = k\left(C_{A_0} \frac{1-X}{1+\varepsilon X}\right)^2 - k_{-B}\left(C_{A_0} \frac{0.5X}{1+\varepsilon X}\right)$$

$$-r_A = 2\left(4 \frac{1-X}{1+0.5X}\right)^2 - 4\left(4 \frac{0.5X}{1-0.5X}\right)$$

$$-r_A = 2\left(16\frac{(1-X)^2}{(1-0.5X)^2}\right) - 4\left(2\frac{X}{1-0.5X}\right)$$

$$-r_A = 32\frac{(1-X)^2}{(1-0.5X)^2} - 8\frac{X}{1-0.5X}$$

$$-r_A = \frac{32(1-X)^2 - 8X(1-0.5X)}{(1-0.5x)^2}$$

$$-r_A = \frac{32(1-2x+x^2) - 8x + 4x^2}{(1-0.5x)^2}$$

$$-r_A = \frac{32 - 64x + 32x^2 - 8x + 4x^2}{(1-0.5x)^2}$$

$$-r_A = \frac{36x^2 - 72X + 32}{(1-0.5x)^2}$$

4.

$$-r_A = k_A[A]^2 - k_{-B}[B]$$

$$k_A = 2\frac{dm^6}{mol^2.s}, \quad k_{-B} = 4\frac{dm^3}{mol.s}$$

$$[A] = C_{A_0}(1-X)$$

$$[B] = C_{A_0} \times 0.5 \times X$$

$$-r_A = k_A((C_{A0}(1-X))^2 - k_{-B}((C_{A0} \times 0.5 \times X)$$

$$-r_A = 2(4.0(1-X))^2 - 4(4 \times 0.5 \times X)$$

$$-r_A = 2(16(1-X)^2] - 4(2X)$$

$$-r_A = 32(1-2X+X^2) - 8X$$

$$-r_A = 32 - 64X + 32X^2 - 8X$$

$$\boxed{-r_A = 32X^2 - 72X + 32}$$

Q4-2$_A$ i>clicker. Go to the Web site (*http://www.umich.edu/~elements/6e/04chap/iclicker_ch4_q1.html*) and view at leave five i>clicker questions. Choose one that could be used as is, or a variation thereof, to be included on the next exam. You also could consider the opposite case, explaining why the question should *not* be on the next exam. In either case, explain your reasoning.

Chapter 4: Stoichiometry

Self Test: I Clicker questions - Multiple Choice Questions

i>clicker Question: Ch4

8) A liquid stream containing 1 mol/dm^3 of A, 2 mol/dm^3 of B, and 0.5 mol/dm^3 of C enters a reactor. The irreversible reaction

$$A + B \rightarrow C$$

proceeds to a conversion of A of 75%. What is the outlet concentration of C?

A) 0.75 mol/dm^3

B) 1.25 mol/dm^3

C) 2 mol/dm^3

D) 2.5 mol/dm^3

$$C_{A_0} = 1 \frac{mol}{dm^3}$$

$$C_{B_0} = \frac{2\,mol}{dm^3}$$

$$C_{C_0} = 0.5 \frac{mol}{dm^3}$$

$$\Delta C_C = \left(\frac{\text{moles (produced}}{\text{moles A reacted}}\right) \times \left(C_{A_0} \cdot X_A\right)$$

$$\Delta C_C = 1 \times \left(1 \frac{mol}{dm^3} \cdot 0.75\right) = 0.75 \frac{mol}{dm^3}$$

$$C_C = C_{C_0} + \Delta C_C$$

$$C_C = 0.5\,\frac{mol}{dm^3} + 0.75\,\frac{mol}{dm^3} = 1.25\,\frac{mol}{dm^3}$$

$\boxed{B}$ 1.25 mol/dm3.

⇒ I chose this question for next exame because it teaches concept of stoichiometic table construction.

⇒ It effectively tests whether a student can distinguish between initial concentration and concentration changes.

⇒ Many times students mistakenly assume the outlet concentration is simply $C_{A_0} X_A (0.75)$, forgetting to account for the species already present in the feed.

⇒ Efficiency: It is an ideal question that ensures students understand liquid phase stoichiometry before moving on to more complex gas-phase systems with expansion factors (ε)

Wolfram and Python

1. Vary Θ_B and observe the change in reaction rate. Go to the extremes and explain what is causing the curve to change the way it does.
2. Vary the parameters and list the parameters that have the greatest and the least effect on the reaction rate.
3. Write a set of conclusions about the experiments you carried out by varying the sliders in parts (i) through (iii).

   Polymath
4. Using the molar flow rate of SO_2 (A) of 3 mol/h and Figure E4-4.1, calculate the fluidized bed (i.e., CSTR catalyst weight) necessary for 40% conversion.
5. Next, consider the entering flow rate of SO_2 is 1000 mol/h. Plot $(F_{A0}/-r_A)$ as a function of X to determine the fluidized bed catalyst weight necessary to achieve 30% conversion, and 99% of the equilibrium conversion, that is, X = 0.99 X_e.

The SO_2 oxidation discussed in Exa[illegible] is to be carried out over a solid platinum catalyst. As with almost all gas–solid catalytic reactions, the catalytic rate law is expressed in terms of partial pressures instead of concentrations. The rate law for this SO_2 oxidation was found experimentally to be[1]

$$-r'_{SO_2} = \frac{k\left[P_{SO_2}\sqrt{P_{O_2}} - \frac{P_{SO_3}}{K_P}\right]}{\left(1+\sqrt{P_{O_2}\,K_{O_2}}+P_{SO_3}K_{SO_3}\right)^2}\text{, mol } SO_2\text{oxidized}/\text{(h)}/\text{(g-cat)} \qquad \text{(E4-4.1)}$$

[1] O. A. Uychara and K. M. Watson, *Ind. Engrg. Chem*, 35, 541.

where P_i (kPa, bar, or atm) is the partial pressure of species *i*.

The reaction is to be carried out isothermally at 400°C. At this temperature, the rate constant k per gram of catalyst (g-cat), the adsorption constants for O_2 (K_{O_2}) and SO_2 (K_{SO_2}), and the pressure equilibrium constant, K_P, were experimentally found to be

$k = 9.7$ mol SO_2/atm$^{3/2}$/h/g-cat

$$K_{O_2} = 38.5\,\text{atm}^{-1},\ K_{SO_3} = 42.5\,\text{atm}^{-1},\ \text{and } K_P = 930\,\text{atm}^{-1/2}$$

The total pressure and the feed composition (e.g., 28% SO_2) are the same as in Example 4-3. Consequently, the entering partial pressure of SO_2 is 4.1 atm. There is virtually no pressure drop in this reactor.

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Code

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import fsolve

# ----------------------------
# GIVEN PARAMETERS
# ----------------------------
thetaB = 0.54
epsilon = -0.14
k = 9.7
KO2 = 38.5
KSO3 = 42.5
KP = 930
PSO20 = 4.1

# ----------------------------
# Partial pressures
# ----------------------------
def partial_pressures(X):
    PSO2 = PSO20*(1-X)/(1+epsilon*X)
    PSO3 = PSO20*X/(1+epsilon*X)
    PO2  = PSO20*(thetaB - X/2)/(1+epsilon*X)
    return PSO2, PSO3, PO2

# ----------------------------
# Rate expression
# ----------------------------
def rate(X):
    PSO2, PSO3, PO2 = partial_pressures(X)
    numerator = k*(PSO2*np.sqrt(PO2) - PSO3/KP)
    denominator = (1 + np.sqrt(PO2*KO2) + PSO3*KSO3)**2
    return numerator/denominator

# ----------------------------
# 1 EQUILIBRIUM CONVERSION
# ----------------------------
Xe = fsolve(rate, 0.5)[0]
print("Equilibrium Conversion Xe =", round(Xe,4))
```

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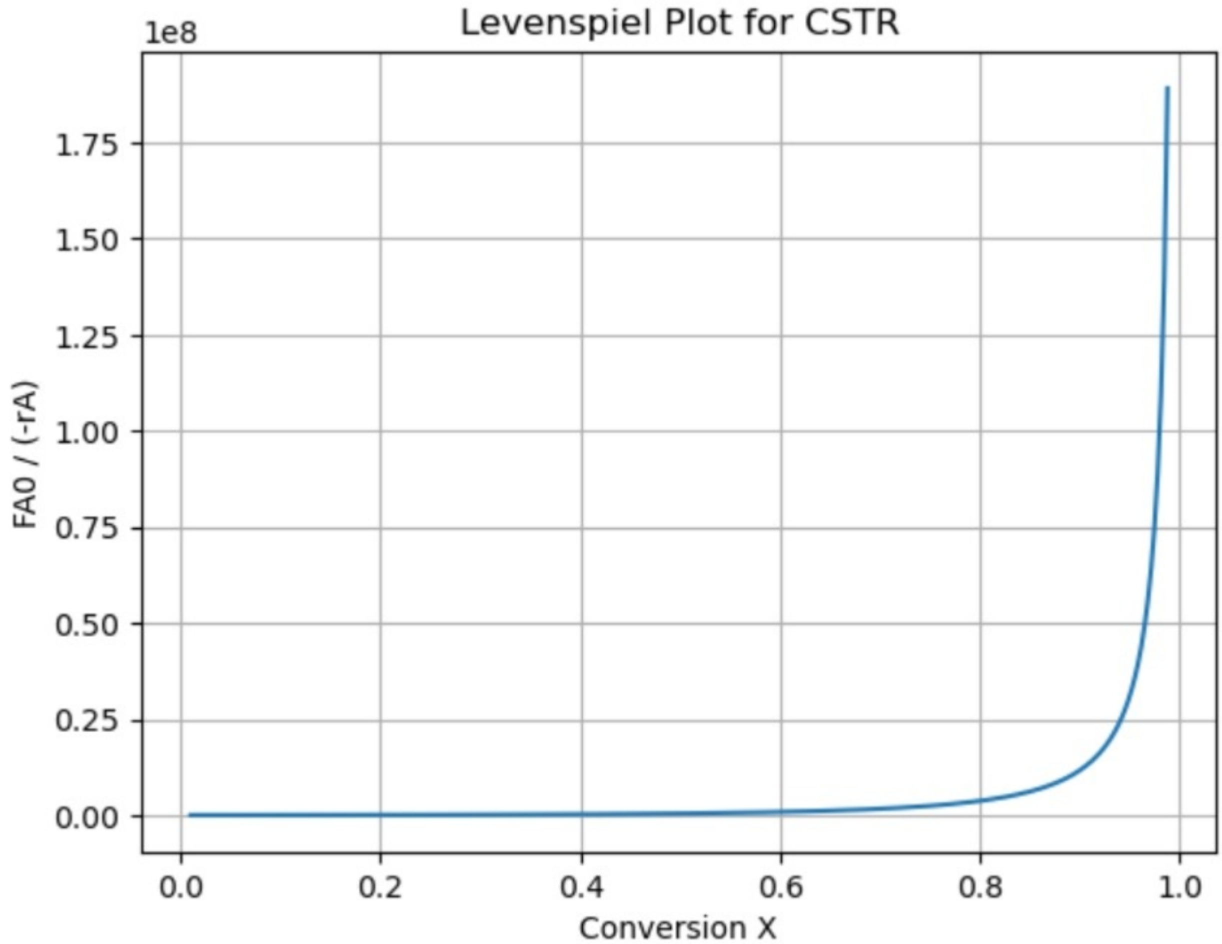
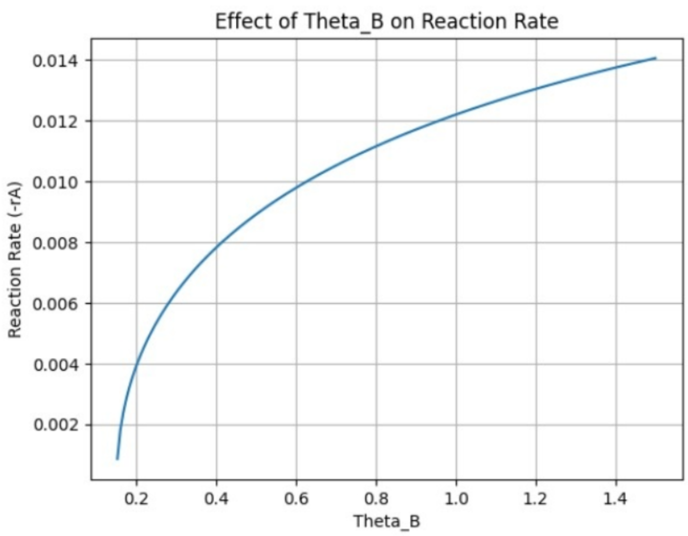
Code

```
# ----------------------------
# 1 EQUILIBRIUM CONVERSION
# ----------------------------
Xe = fsolve(rate, 0.5)[0]
print("Equilibrium Conversion Xe =", round(Xe,4))

# ----------------------------
# 4 CSTR weight (FA0=3 mol/h, X=0.40)
# ----------------------------
FA0_1 = 3
X1 = 0.40
r1 = rate(X1)
W1 = FA0_1*X1/r1
print("CSTR weight for 40% conversion =", round(W1,2), "g-cat")

# ----------------------------
# 5 FA0 = 1000 mol/h
# ----------------------------
FA0_2 = 1000

X_vals = np.linspace(0.01, 0.99*Xe, 300)
FA_over_r = FA0_2 / rate(X_vals)

plt.plot(X_vals, FA_over_r)
plt.xlabel("Conversion X")
plt.ylabel("FA0 / (-rA)")
plt.title("Levenspiel Plot for CSTR")
plt.grid()
plt.show()

# Weight at 30%
X30 = 0.30
W30 = FA0_2*X30/rate(X30)
print("Weight for 30% conversion =", round(W30,2), "g-cat")

# Weight at 99% equilibrium
X99 = 0.99*Xe
W99 = FA0_2*X99/rate(X99)
print("Weight for 99% of equilibrium =", round(W99,2), "g-cat")
```

```
Equilibrium Conversion Xe = 0.9976
CSTR weight for 40% conversion = 265.24 g-cat
```




```
Weight for 30% conversion = 32316.89 g-cat
Weight for 99% of equilibrium = 186776841.81 g-cat
```



1. Rate increases as θ_B increases

At low θ_B : O_2 is limiting → rate small

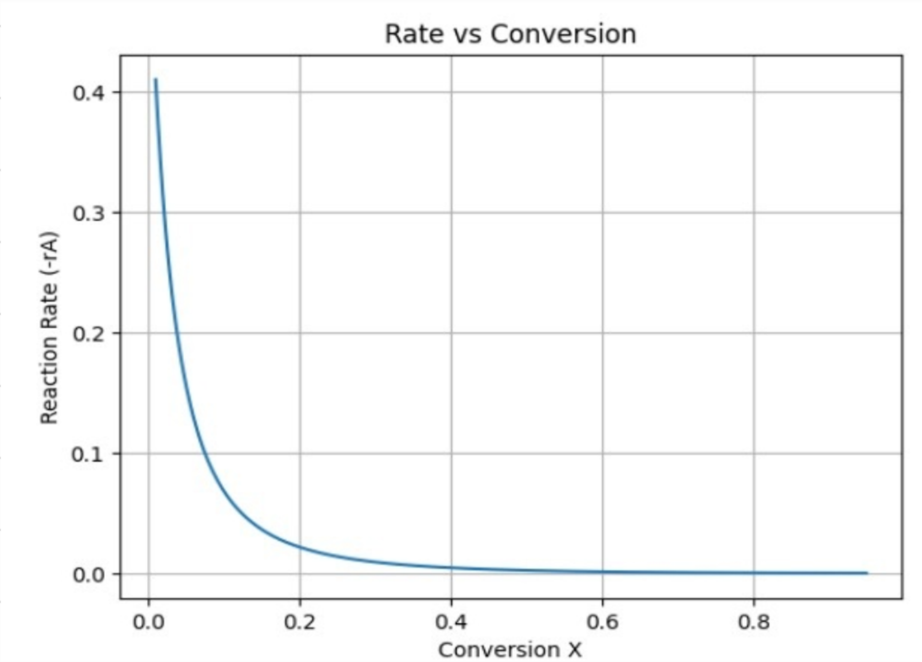
As θ_B increases : more O_2 available →

rate increases

At very high values the increase slows due to absorption denominator effects.

(2) Parameter sensitivity (20% increase)

Parameter	Sensitivity
K	+0.2 (strongest +ve effect)
KSO_3	-0.2715 (largest -ve effect)
KO_2	-0.0236 (small effect)
KP	0.0001 (negligible effect)



(3.) Rate is highest near $X = 0$

Rate decreases sharply as X increases

Near equilibrium, rate $\rightarrow 0$

Reason:

SO_2 decreases

SO_3 increases

Reverse reaction term grows

Driving force goes to zero.

4. CSTR weight

$W = 265.24$ g-cat

$F_{A0} = 3 \frac{mol}{h}$ $X = 0.40$

5. Large feed case ($F_{A0} = 1000 \frac{mol}{h}$)

$X_e = 0.9976$

(a) For 30% conversion $W = 32,316.89$ g-cat

(b) For 99% of equilibrium conversion;

$W = 1.87 \times 10^8$ g-cat

Extremely large because $r_A \rightarrow 0$ near equilibrium.

So reactor size → ∞.

(b) Example 4-5: Equilibrium Conversion

1. Using Figure E4-5.1 and the methods in Chapter 2, estimate the PFR volume for 40% conversion for a molar flow rate of A of 5 dm^3/min.
2. Using Figure E4-5.1 and the methods in Chapter 2, estimate the time to achieve 40% conversion in a batch reactor when the initial concentration is 0.05 mol/dm^3.

   Wolfram and Python
3. What values of K_C and C_{A0} cause X_{ef} to be the farthest away from X_{eb}?
4. What values of K_C and C_{A0} will cause X_{eb} and X_{ef} to be the closest together?
5. Observe the plot of the ratio of $\left(\frac{X_{eb}}{X_{er}}\right)$ as a function of y_{A0}. Vary the values of K_C and C_{T0}. What conclusions can you draw?
6. Write a set of conclusions from your experiment (i) through (vi).

1. **Batch system—constant volume, $V = V_0$.**

TABLE E4-5.1 STOICHIOMETRIC TABLE FOR BATCH REACTOR

Species	*Symbol*	*Initially*	*Change*	*Remaining*
N_2O_4	A	N_{A0}	$-N_{A0}X$	$N_A = N_{A0}(1 - X)$
NO_2	B	0	$+2N_{A0}X$	$N_B = 2N_{A0}X$
		$N_{T0} = N_{A0}$		$N_T = N_{T0} + N_{A0}X$

(1) PFR Volume at $X = 0.40$

$$V = F_{A0} \int_0^X \frac{dX}{-r_A} \qquad F_{A0} = 5\,\frac{dm^3}{min}$$

X	$1/-r_A$
0.0	0.5
0.2	1
0.4	2

Estimate Area $(0 \rightarrow 0.4)$

$0 \rightarrow 0.2$

$$A_1 = \frac{(0.5+1.0)}{2}(0.2) = 0.15$$

$0.2 \rightarrow 0.4$

$$A_2 = \frac{(1.0+2.0)}{2}(0.2) = 0.30$$

Total Area

$$\int_0^{0.4} \frac{dX}{-r_A} = 0.15 + 0.30 = 0.45$$

$$V = F_{A_0} \times A$$

$$V = 5 \times 0.45 = \boxed{2.25\ dm^3}$$

$$\boxed{V_{PFR} \simeq 2.3\ dm^3}$$

(2): For Batch reactor

$$t = \int_0^{X} \frac{dX}{-r_A}.$$

$$t = \int_0^{0.4} \frac{dX}{-r_A}$$

$0 \to 0.2 \quad A_1 = \frac{(0.5+1.0)}{2}(0.2) = 0.15$

$0.2 \to 0.4$

$$A_2 = \frac{(1.0+2.0)}{2}(0.2) = 0.30$$

$t = 0.15 + 0.30 = 0.45$ min

$$\boxed{t_{40\%} \approx 0.45 \text{min}}$$

(3) The difference between X_{ef} and X_{eb} is greatest when

$K_c \approx 1$ and C_{A_0} is large

(4.) $K_c >> 1$

$X_{eb} \approx 1$ & $X_{ef} \approx 1$ difference ≈ 0

$K_c << 1$

$X_{eb} \approx 0$ & $X_{ef} \approx 0$ difference ≈ 0

when $C_{A_0} \to 0$

$X_{eb} \approx X_{ef}$

Thus X_{eb} and X_{ef} are closes when

$K_c << 1$ or $K_c >> 1$ and C_{A_0} is small

(5.) The difference between X_{eb} and X_{ef} is greatest when $\frac{K_C}{C_{T_0}} \approx 1$

(6.) Overall conclusions

(1) Equilibrium conversion depends on both K_C and initial concentration.

(2) When reaction does not change total moles

$$X_{eb} = X_{ef} \quad (\Delta\nu = 0)$$

for all values of K_C and C_{T_0}

(3) When $\Delta\nu \neq 0$, equilibrium conversion differ between batch and flow systems.

(4) The difference between X_{eb} and X_{ef} increases as:

→ The change in moles ($\Delta\nu$) increases
→ K_C / C_{T_0} approaches 1

(5) $\frac{K_C}{C_{T_0}} << 1$ (unfavorable reaction)

$\frac{K_C}{C_{T_0}} >> 1$ (strongly favorable reaction)

C_{T_0} is very small (dilute system)

⑥ Flow reactors are more sensitive to mole-number changes because concentration changes continuously along the reactor.